

Introduction Of Data Pipeline

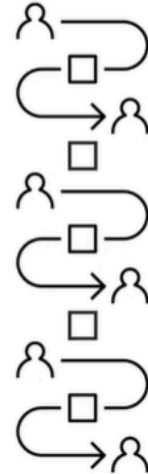
Sure! Let's talk about data pipelines in simple terms.

A data pipeline is like a conveyor belt in a factory. Imagine you have a series of processes lined up, where each process does a specific job. Just like a conveyor belt moves items from one station to the next, a data pipeline moves data from one place to another. For example, if you want to send a box of toys from one room to another, you could have friends passing the box along the line. Each friend takes the box and hands it to the next friend until it reaches the final destination. In a data pipeline, data packets are passed along from one process to another, transforming or moving the data as needed.

Now, there are two important concepts to understand when it comes to data pipelines: latency and throughput. Latency is like the time it takes for a single box to travel from the start of the conveyor belt to the end. If one friend is slow at passing the box, it will take longer for the box to reach the end. Throughput, on the other hand, is about how many boxes can be passed along the conveyor belt in a certain amount of time. If your friends can pass bigger boxes faster, you can move more toys in less time!

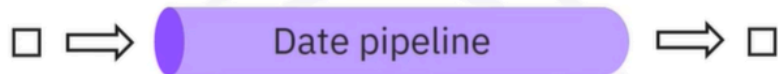
What is a pipeline?

- Series of connected processes
- The output of one process is the input of the next
- For example, take a box, pass it to your neighbor, and so on, until the box arrives at the end of the line
- Mass production: Parts pass along conveyor belts between manufacturing stages



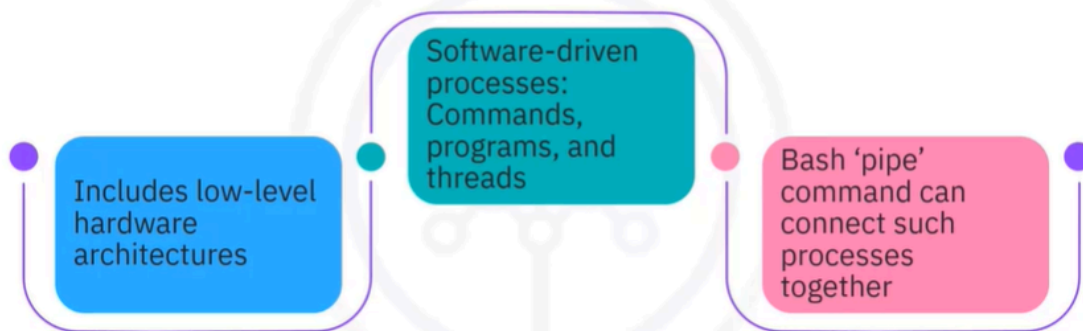
- The concept of a pipeline applies broadly to any set of processes that are connected to each other sequentially. This means that the output of one process is passed along as input to the next process in a chain. For example, one way to move boxes from one place to another is to have a chain of friends, each passing the boxes one by one up the line to the nearest friend. Each friend is a processor whose function is identical, get a box, pass a box. Mass production is similar, but transformations may differ from stage to stage.

What is a data pipeline?



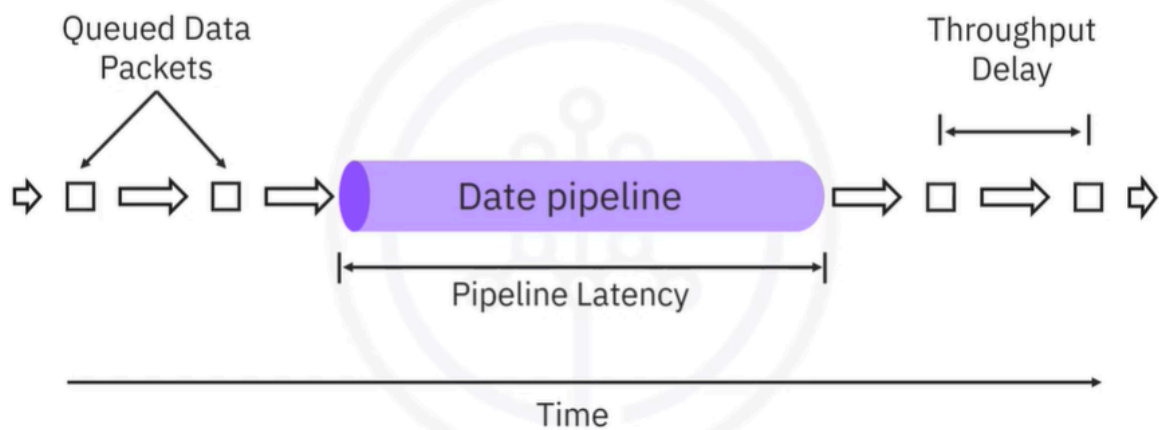
- Purpose: Move data from one place or form to another
 - Any system which extracts, transforms, and loads data
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- Data pipelines are pipelines that specifically move or modify data. The purpose of a data pipeline is to move data from one place or form to another. A data pipeline is a system which extracts data and passes it along to optional transformation stages for final loading.

What is a data pipeline?



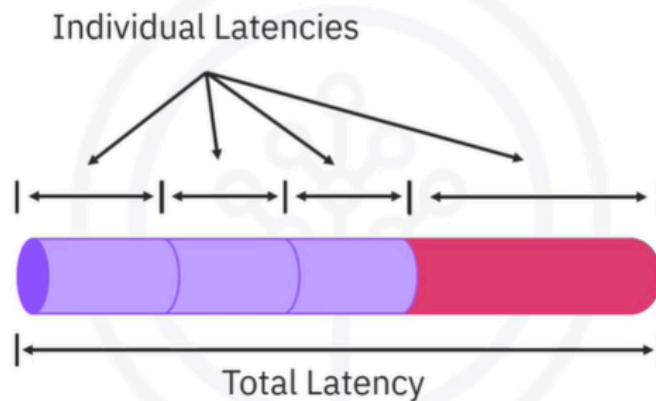
- This includes low-level hardware architectures, but our focus here is on data pipelines as architectures driven by software processes such as commands, programs, and processing threads. The simple bash pipe command in Linux can be used as the glue that connects such processes together.

Packet flow through a pipeline



- We can think of data flowing through a pipeline in the form of data packets. A term which we will use to broadly refer to units of data. Packets can range in size from a single record or event to large collections of data. Here we have data packets queued for ingestion to the pipeline. The length of the data pipeline represents the time it takes for a single packet to traverse the pipeline. The arrows between packets represent the throughput delays or the times between successive packet arrivals.

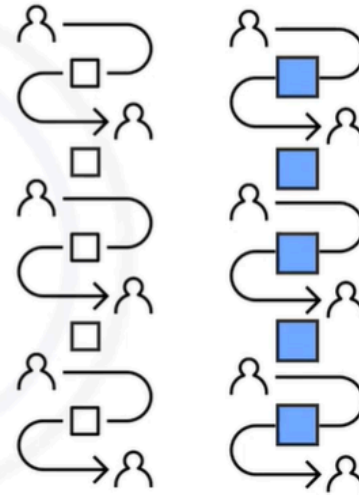
Data pipeline performance: Latency



- You have just been introduced to two key performance considerations regarding data pipelines. The first is latency, which is the total time it takes for a single packet of data to pass through the pipeline. Equivalently, latency is the sum of the individual time spent during each processing stage within the pipeline. Thus, overall latency is limited by the slowest process in the pipeline. For example, no matter how fast your internet service is, the time it takes to load a web page will be decided by the server speed.

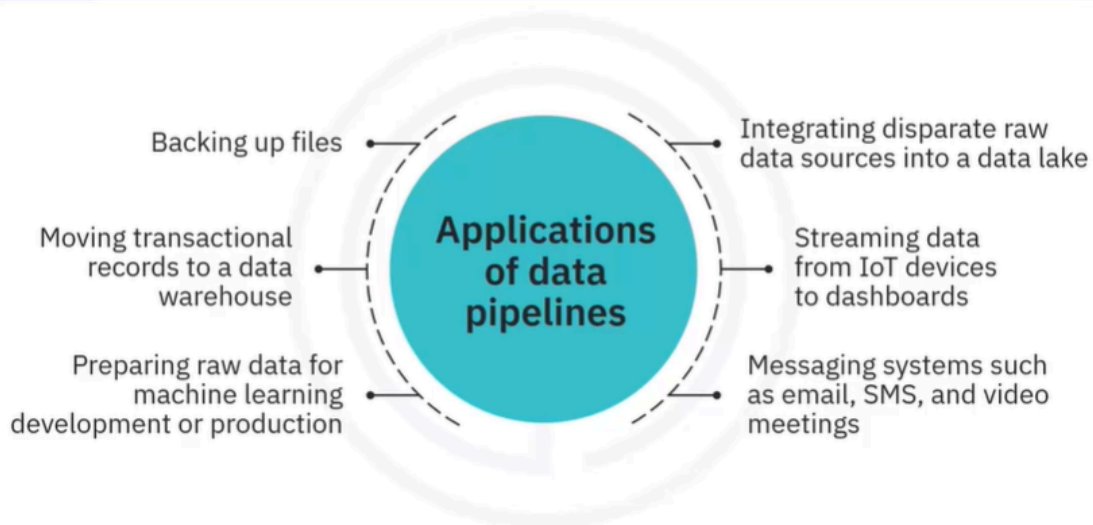
Pipeline performance: Throughput

- How much data can be fed through the pipe per unit of time?
- Larger data packets per time unit equals greater throughput



- The second performance consideration is called throughput. It refers to how much data can be fed through the pipeline per unit of time. Processing larger packets per unit of time increases throughput. Coming back to our example of having a chain of friends passing boxes from one to another, we can see in the right image within limits, that passing bigger boxes can increase productivity.

Use cases



- Let's list a few of the applications of data pipelines from the multitude of use cases. The simplest pipeline has no transformations and is used as file backups, integrating disparate raw data sources into a data lake, moving transactional records to a data warehouse, streaming data from IoT devices to make information available in the form of dashboards or learning systems, preparing raw data for machine learning development or production, and message sending and receiving, such as with email, SMS, or online video meetings.

Recap

In this video, you learned that:

- Data pipelines move data from one place or form to another
- Data flows through pipelines as a series of data packets
- Latency and throughput are key design considerations for data pipelines
- Data pipeline use cases range from simple copy-and-paste to online video meetings

How does latency affect data pipelines?

Latency affects data pipelines by determining how long it takes for a single piece of data (or data packet) to travel through the entire pipeline. Here are some key points to understand:

- **Total Time:** Latency is the total time from when data enters the pipeline until it exits. If any stage in the pipeline is slow, it increases the overall latency.
- **Slowest Process:** The latency of the entire pipeline is limited by the slowest process. For example, if one step takes longer than the others, it will delay the entire pipeline.

- **User Experience:** High latency can lead to delays in data availability, which can affect applications that rely on real-time data, such as online services or dashboards.
- **Optimization:** Reducing latency is crucial for improving the performance of data pipelines. This can involve optimizing processes, using faster hardware, or streamlining data flow.

In summary, lower latency means faster data processing and better performance for applications that depend on timely data.